

Lung-Region Focused Deep Learning Framework for Multi-Class Pneumonia Detection Using EfficientNetV2 and Confidence-Aware Decision Support

Dissertation Report

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Master of Science in Data Science

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DECLARATION

I, the undersigned, hereby declare that the project report entitled “Lung-Region Focused Deep Learning Framework for Multi-Class Pneumonia Detection Using EfficientNetV2 and Confidence-Aware Decision Support”, submitted for partial fulfilment of the requirements for the award of the degree of Master of Science in Data Science, University of Kerala, is a Bonafide work done by me under the supervision of Rahul D, Assistant Professor, Department of Futures Studies.

This submission represents my original work and where ideas or words of others have been included, they have been duly acknowledged, cited, and referenced. I further declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated any data, source, or result in this report.

I also understand that any violation of the above will lead to disciplinary action by the Institute or University and may invoke penal action from the concerned authorities. This report has not been previously submitted for the award of any degree, diploma, or title at any other University or Institution.

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CERTIFICATE

This is to certify that the report entitled “**Lung-Region Focused Deep Learning Framework for Multi-Class Pneumonia Detection Using EfficientNetV2 and Confidence-Aware Decision Support**”, submitted by **RIFA FATHIMA** to the University of Kerala in partial fulfilment of the requirements for the award of the Degree of **Master of Science in Data Science, Department of Futures Studies**, is a Bonafide record of the project work carried out by her under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Pneumonia remains as one of the leading causes of respiratory illness worldwide, requiring timely and accurate diagnosis to reduce morbidity and mortality. Chest X-ray imaging is widely used as a primary diagnostic tool due to its accessibility and cost-effectiveness; however, manual interpretation is time-consuming and subject to inter-observer variability. Recent advances in deep learning have enabled automated analysis of radiographic images, offering potential support for clinical decision-making.

This project presents a lung-region focused deep learning framework for multi-class pneumonia detection from chest X-ray images. The proposed system integrates a supervised U-Net lung segmentation model, trained exclusively on pixel-level lung masks provided with the COVID-19 Radiography Dataset. A key design principle is that these provided masks are used solely to supervise U-Net training; during classification preprocessing, the trained U-Net generates lung masks autonomously on raw, unmasked X-ray images. This design simulates real-world deployment conditions where ground-truth masks would be unavailable, and validates the segmentation model's generalisation beyond its training supervision.

EfficientNetV2-B0 is employed as the primary classification backbone due to its efficient compound scaling strategy and superior training stability. A two-stage transfer learning approach is adopted, consisting of classifier head training on a frozen backbone, followed by selective fine-tuning of the upper backbone layers. The model classifies chest X-ray images into four diagnostic categories: Normal, COVID-19 Pneumonia, Viral Pneumonia, and Bacterial Pneumonia.

Beyond classification, the framework incorporates confidence calibration via temperature scaling, occlusion sensitivity explainability analysis, t-SNE feature visualisation using penultimate-layer embeddings, and a confidence-based clinical decision support mechanism. The model achieved an overall test accuracy of 78.3% and a macro F1-score of 0.78 across all four classes. ROC AUC scores ranged from 0.93 to 1.00, demonstrating strong discriminative capability. Real-world inference testing demonstrated clean lung segmentation on unseen X-ray images and appropriate low-confidence flagging for ambiguous cases, confirming the clinical safety orientation of the system.

The proposed framework emphasises reliability, interpretability, and clinical applicability, highlighting the importance of integrating explainable AI and confidence-aware decision mechanisms in medical imaging systems.

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1. INTRODUCTION

Respiratory diseases, particularly pneumonia, continue to pose a significant global health burden. Pneumonia is an inflammatory condition of the lung parenchyma caused by bacterial, viral, or fungal infections, and its early detection is critical for effective treatment and the prevention of severe complications. Chest radiography remains the most commonly used imaging modality for pneumonia diagnosis due to its affordability, rapid acquisition, and widespread availability across clinical settings.

However, interpreting chest X-ray images requires specialised radiological expertise and is subject to human fatigue, cognitive bias, and inter-observer variability. These limitations create both a need and an opportunity for automated, reliable diagnostic support systems grounded in artificial intelligence.

Deep learning, particularly convolutional neural networks, has demonstrated remarkable performance in medical image classification tasks. However, many published approaches adopt end-to-end classification without addressing the reliability, interpretability, or segmentation-generalisation aspects of the model. Models trained on raw chest X-rays may inadvertently learn spurious correlations from irrelevant background features such as image borders, medical device artifacts, or text annotations, leading to shortcut learning and reduced clinical trustworthiness.

Lung-region focused preprocessing has emerged as an effective strategy to guide model attention toward anatomically relevant areas. U-Net architectures, originally developed for biomedical image segmentation, enable pixel-wise binary segmentation of pulmonary regions from chest radiographs. A key design decision in this work is that the pre-provided lung masks from the COVID-19 Radiography Dataset are used only to supervise U-Net training. During classification preprocessing, the trained U-Net generates masks autonomously on raw, unmasked images, thereby validating the segmentation model’s generalisation capability and ensuring the classification pipeline is not dependent on manually annotated masks at inference time.

EfficientNetV2-B0 is employed as the classification backbone due to its compound scaling strategy, improved training stability over its predecessor, and consistently strong performance on medical imaging tasks. A two-stage transfer learning approach with class-weight balancing, controlled augmentation, and temperature-scaled confidence calibration ensures both accuracy and prediction reliability.

This project proposes an integrated framework combining supervised lung segmentation, image enhancement, EfficientNetV2-based multi-class classification, confidence calibration, and explainability analysis. The system is designed to prioritise reliability and clinical safety alongside classification performance, and is validated through comprehensive quantitative evaluation and real-world inference testing.

2. LITERATURE REVIEW

The application of deep learning to pneumonia detection from chest X-ray images has attracted significant research interest over the past decade. Early approaches focused primarily on improving classification accuracy; more recent studies emphasise preprocessing, segmentation, explainability, and reliability to enhance clinical applicability. The reviewed literature is organised into five conceptual areas relevant to the proposed work.

Base Paper: Multi-Class Pneumonia Detection Using Transfer Learning

The base paper by Ibrahim et al. (2021) presents a deep learning framework for classifying chest X-ray images into Normal, Bacterial Pneumonia, Viral Pneumonia, and COVID-19 Pneumonia using AlexNet pretrained on large-scale image datasets and fine-tuned for pneumonia classification. The model achieved approximately 92% accuracy in multi-class classification on the COVID-19 Radiography Dataset.

Despite this reported performance, several limitations were identified: AlexNet is a shallow architecture compared to modern convolutional neural networks; no lung segmentation is applied to guide model attention; no image enhancement is incorporated; and explainability and confidence calibration are entirely absent. Crucially, the model may rely on background artifacts rather than clinically meaningful features. These limitations collectively motivate all enhancements introduced in the present work.

1. Deep Learning Architectures for Pneumonia Detection

Rajpurkar et al. (2017) introduced CheXNet, a DenseNet-121-based model achieving radiologist-level pneumonia detection with an AUC of approximately 0.76 on the NIH ChestX-ray14 dataset, establishing the feasibility of automated chest X-ray interpretation at clinical-grade performance. Tan and Le (2019) proposed the EfficientNet family using compound scaling across depth, width, and resolution dimensions simultaneously, achieving state-of-the-art performance across multiple benchmarks with high computational efficiency. Banerjee and Monir (2024) subsequently demonstrated that EfficientNetV2 outperforms EfficientNetV1 in medical imaging tasks, offering faster convergence, improved training stability, and reduced sensitivity to hyperparameter choices. Chowdhury et al. (2020) achieved 90–95% accuracy in distinguishing COVID-19, viral pneumonia, and normal cases, while noting inherent challenges in differentiating pathological subtypes due to overlapping radiographic presentations.

2. Lung Segmentation and Region-Focused Learning

Ronneberger et al. introduced U-Net, a fully convolutional architecture designed for biomedical image segmentation featuring an encoder-decoder structure with skip connections that preserve spatial detail across resolution levels. This architecture has become the standard for medical image segmentation due to its data efficiency and strong performance with limited training samples.

Haghanifar et al. (2022) demonstrated that lung-focused preprocessing significantly improves COVID-19 classification accuracy by removing irrelevant background structures and directing model attention exclusively toward pulmonary regions. The present work extends this principle by using the provided dataset masks solely for U-Net training supervision, then applying the trained model to generate masks on raw images, constituting a deliberate architectural decision that tests cross-image generalisation.

3. Image Enhancement and Preprocessing

Altan and Narlı (2022) demonstrated that contrast enhancement techniques significantly improve COVID-19 detection accuracy by enhancing ground-glass opacities and pulmonary infiltrates that are frequently subtle and low-contrast in standard radiographic images. The proposed framework applies brightness enhancement prior to classification to improve the visibility of such pathological features, contributing to more robust feature extraction by the classification model.

4. Explainability and Visualisation Methods

Simonyan et al. (2014) introduced occlusion sensitivity analysis as a model-agnostic approach to visualise convolutional neural network decision-making by systematically masking input regions and measuring the resulting change in prediction confidence. While Gradient-weighted Class Activation Mapping is more commonly adopted, its effectiveness diminishes in very deep architectures such as EfficientNetV2 due to gradient diffusion through many layers. Occlusion-based explainability provides stable, interpretable visual explanations and is particularly well-suited for deep architectures in medical imaging applications.

5. Confidence Calibration and Reliability in Medical AI

Guo et al. (2017) systematically analysed neural network calibration and proposed temperature scaling as an effective post-hoc calibration technique that improves the alignment between predicted confidence scores and true class likelihoods without modifying classification accuracy. In medical imaging, poorly calibrated probability estimates may lead to overconfident incorrect predictions and unsafe clinical decisions. Moses (2021) emphasised the necessity of confidence-aware systems in clinical AI pipelines, particularly for high-stakes diagnostic tasks. The present work implements temperature scaling and provides a decision support mechanism based on calibrated confidence thresholds, ensuring that uncertain predictions are flagged for expert review rather than accepted automatically.

Research Gaps Addressed by the Proposed Work

Limitations in Existing Studies	Advancements in the Proposed Work
Focus solely on accuracy without reliability assessment	Temperature-scaled confidence calibration for reliable probability estimates
Models learn spurious correlations from background artifacts, leading to shortcut learning	U-Net lung segmentation restricts classifier input to anatomically relevant regions
Pre-provided masks used directly without validating segmentation generalisation	Masks used only for U-Net supervision; U-Net generates masks autonomously on unmasked images
Shallow or computationally inefficient CNN architectures	EfficientNetV2-B0 with two-stage transfer learning for stable and efficient convergence
Subtle pneumonia patterns missed due to low image contrast	Brightness enhancement applied to improve visibility of faint pathological features
Lack of explainability reduces clinical trust	Occlusion sensitivity maps and t-SNE visualisation provide model transparency
No reliability-based clinical routing	Three-tier confidence-based decision support routing system

3. PROBLEM STATEMENT

Pneumonia remains a major global health concern, affecting millions of individuals annually and contributing significantly to morbidity and mortality across all age groups. Early and accurate diagnosis is essential for effective treatment, appropriate antibiotic or antiviral prescription, and prevention of severe complications including respiratory failure and sepsis.

Chest X-ray imaging is widely adopted as the primary diagnostic tool due to its availability, affordability, and rapid acquisition. However, manual interpretation requires specialised expertise and is subject to significant variability due to radiologist fatigue, experience level, and the inherently subtle nature of early pneumonia findings. These limitations create persistent demand for reliable automated diagnostic support systems.

Recent advances in deep learning have enabled automated pneumonia detection with promising classification accuracy. Despite these advancements, several challenges limit the clinical adoption of existing systems. Many models rely on end-to-end classification without lung-region focusing, potentially learning spurious correlations from irrelevant image regions. This shortcut learning behaviour degrades performance when tested on images from different acquisition settings.

A critical challenge is the absence of segmentation, generalisation and validation in most published work. When pre-provided segmentation masks are used directly as classification inputs, the resulting pipeline cannot be deployed in clinical settings where ground-truth masks are unavailable. The proposed work addresses this by restricting mask usage to U-Net training supervision exclusively, with the trained model generating masks autonomously during inference.

Additionally, deep neural networks frequently produce overconfident probability estimates that do not reflect true diagnostic uncertainty. Most existing systems provide no confidence-aware decision support, no interpretability mechanisms, and no means to flag uncertain predictions for expert radiologist review. This overconfidence is clinically dangerous, as a high-confidence incorrect prediction may be accepted without appropriate scrutiny.

There is therefore a need for a robust, interpretable, and reliability-aware framework that focuses on clinically relevant lung regions, validates segmentation generalisation, provides calibrated confidence estimates, and incorporates explainability analysis to support and augment clinical decision-making rather than replace it.

4. OBJECTIVES OF THE STUDY

The primary goal of this project is to develop a reliable, interpretable, and clinically oriented deep learning framework for multi-class pneumonia detection from chest X-ray images. The specific objectives are:

1. To develop a multi-class pneumonia classification system capable of accurately distinguishing between Normal, Bacterial Pneumonia, Viral Pneumonia, and COVID-19 Pneumonia from chest X-ray images.
2. To implement lung-region segmentation using a U-Net architecture, trained on supervised image-mask pairs, to isolate pulmonary regions and suppress irrelevant background structures during classification preprocessing.
3. To validate U-Net segmentation generalisation by applying the trained model to raw, unmasked X-ray images during classification, simulating deployment conditions where ground-truth masks are unavailable.
4. To apply image enhancement techniques to improve the visibility of subtle pathological features such as ground-glass opacities and pulmonary infiltrates.
5. To employ EfficientNetV2-B0 as a modern and computationally efficient deep learning architecture for accurate feature extraction from chest radiographs.
6. To implement a two-stage transfer learning strategy comprising classifier head training on a frozen backbone followed by selective top-layer fine-tuning for improved domain adaptation.
7. To evaluate model performance using comprehensive quantitative metrics including precision, recall, F1-score, confusion matrix analysis, ROC curves, Precision-Recall curves, and misclassification analysis.
8. To apply temperature scaling for post-hoc confidence calibration to align predicted probability estimates with true diagnostic likelihoods.
9. To incorporate occlusion sensitivity analysis and t-SNE feature visualisation to provide model transparency and support clinical interpretability.
10. To design a confidence-based decision support mechanism that routes predictions into three clinical pathways: automated acceptance for high-confidence predictions, expert radiologist review for moderate-confidence predictions, and advanced imaging recommendation for low-confidence predictions.
11. To validate the complete pipeline through real-world inference testing on unseen chest X-ray images, demonstrating end-to-end functionality and clinical applicability.

5. SYSTEM OVERVIEW AND PROPOSED FRAMEWORK

System Overview

The proposed system is a two-phase end-to-end deep learning framework for pneumonia detection from chest X-ray images. The two phases are designed to be structurally independent, enabling the classification phase to proceed from a persisted intermediate dataset without re-executing the computationally expensive segmentation phase.

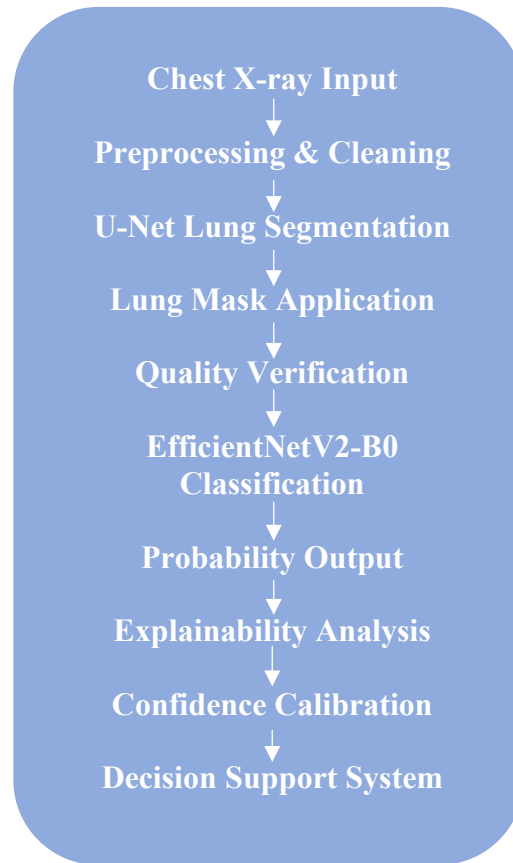
Phase 1 constitutes the segmentation pipeline. A U-Net model is trained using image-mask pairs drawn from the COVID-19 Radiography Dataset, where pixel-level binary lung masks serve as ground-truth supervision. Upon training completion, the U-Net is applied to all raw, unmasked chest X-ray images across training, validation, and test partitions to generate a masked classification dataset. This dataset is persisted to disk, forming the input to Phase 2.

Phase 2 constitutes the classification pipeline. The persisted masked dataset is loaded and used to train an EfficientNetV2-B0 classifier through a two-stage transfer learning procedure. Following classification training, the system evaluates performance through comprehensive quantitative metrics, analyses model reliability through temperature-scaled calibration, visualises learned feature representations through t-SNE, and provides prediction interpretability through occlusion sensitivity mapping. A confidence-based decision support mechanism routes final predictions into appropriate clinical pathways.

Key Design Principles

- **Supervised Segmentation with Generalisation Validation:** The U-Net is trained on provided image-mask pairs and applied to raw unmasked images during classification, validating that the segmentation model generalises beyond its training ground truth.
- **Separation of Segmentation and Classification Phases:** The two phases are executed independently with a persisted intermediate dataset, ensuring reproducibility and enabling resumption of the classification phase without repeating segmentation.
- **Reliability-First Design:** Confidence calibration and three-tier decision support ensure that prediction uncertainty is communicated clearly rather than masked by overconfident probability estimates.
- **Interpretability Integration:** Occlusion sensitivity maps and t-SNE feature visualisation are embedded as core components, not post-hoc additions, ensuring that model transparency is maintained throughout the evaluation pipeline.

Pipeline Workflow



Step	Description
1	Chest X-ray dataset acquired programmatically via Kaggle API
2	Image cleaning, resizing (224×224), class balancing, and stratified splitting (70/15/15)
3	Image-mask pairs collected; U-Net trained under supervised segmentation on provided masks
4	Trained U-Net applied to raw unmasked images to generate masked classification dataset
5	Quality verification: scan for empty masks, compute fallback rate
6	EfficientNetV2-B0 trained: Stage 1 (frozen backbone), Stage 2 (top-30 fine-tuning)
7	Evaluation: classification report, confusion matrix, ROC, PR curves, misclassification analysis
8	t-SNE feature visualisation using penultimate-layer embeddings
9	Temperature scaling calibration; reliability diagram generated
10	Occlusion sensitivity explainability maps generated for selected samples
11	Real-world inference: unseen X-ray uploaded, masked, predicted, and routed through decision support

6. METHODOLOGY AND SYSTEM ARCHITECTURE

1. Data Acquisition and Dataset Description

The dataset used in this study is the COVID-19 Radiography Database, curated by Tawsifur Rahman et al. and publicly available on Kaggle. Dataset acquisition is performed programmatically, enabling fully reproducible retrieval without manual intervention. The dataset comprises posterior-anterior chest X-ray images across four diagnostic categories:

Class	Image Count	Notes
Normal	20,384	Healthy lung appearances
Lung Opacity (Bacterial proxy)	12,024	Mapped to BACTERIAL class
COVID-19 Pneumonia	7,232	Ground-glass opacities, bilateral involvement
Viral Pneumonia	2,690	Diffuse interstitial patterns

A key property of this dataset is that each class folder contains both an images subdirectory and a masks subdirectory providing pixel-level binary lung segmentation masks for every image. These masks are used exclusively to supervise U-Net training and are never used as direct inputs to the classification pipeline.

Note on label mapping: Lung Opacity is mapped to the BACTERIAL class as a clinical proxy for non-COVID, non-viral pneumonia. This mapping is acknowledged as an approximation, since Lung Opacity encompasses multiple aetiologies beyond purely bacterial causes, and is explicitly stated as a limitation of the label schema.

2. Data Preprocessing and Balancing

i. Image-Mask Pair Collection

Image-mask pairs for U-Net training are identified and catalogued prior to any dataset reorganisation. The original dataset maintains a per-class folder structure in which X-ray images and their corresponding binary lung masks are stored in separate subdirectories. Since subsequent preprocessing consolidates images into a unified class-labelled directory structure, the image-to-mask correspondence must be established beforehand to preserve the pairing required for supervised segmentation training.

ii. Dataset Organisation and Class Mapping

Images are copied from their original class-specific source folders into a unified directory structure organised by class label. A class mapping schema accommodates all known folder name variants present in the dataset, including alternative spellings and abbreviations. Where multiple source

folders map to the same target class, filename collision handling ensures no image is overwritten during consolidation.

iii. Data Cleaning

Corrupted, unreadable, or truncated images are identified and removed using image verification prior to balancing. This step prevents training instability arising from malformed inputs and ensures that all images used in subsequent stages are structurally valid.

iv. Image Resizing

All images are resized to 224×224 pixels to match the input requirements of EfficientNetV2-B0:

$$I_{resized} = \text{resize}(I_{original}, 224 \times 224)$$

Although chest X-rays are inherently grayscale, they are represented across three identical channels to match the three-channel input expected by ImageNet-pretrained weights. This enables transfer learning and substantially reduces the training data required to achieve competitive performance.

Pixel Values are normalised to the range $[0,1]$ to stabilise gradient updates during training:

$$I_{norm} = \frac{I}{255}$$

Normalisation improves convergence speed and training stability.

v. Dataset Balancing

The raw class distribution is highly imbalanced, with Normal comprising 20,384 images and Viral Pneumonia comprising only 2,690. Controlled resampling is applied: minority classes are augmented up to the target count using a single controlled augmentation pass per image, and the dominant class is undersampled. A single augmentation pass per image is applied to avoid disproportionate over-representation of any class. Augmentation operations include horizontal flipping, small rotations, and contrast and brightness perturbation. Following balancing, all four classes contain 7,000 images each, confirmed by class weight computation returning equal weights of 1.0000 for all classes.

vi. Dataset Splitting

The balanced dataset is partitioned into training (70%), validation (15%), and test (15%) subsets using stratified splitting to preserve class distribution across all three partitions. This produces 7,000 training images, 1,500 validation images, and 1,500 test images per class, for a total of 28,000 training, 6,000 validation, and 6,000 test images.

3. Phase 1 — U-Net Lung Segmentation

i. Design Rationale

The COVID-19 Radiography Dataset provides high-quality lung masks generated by a validated clinical pipeline. A naive approach would apply these masks directly to classification inputs;

however, this creates a pipeline that cannot generalise to clinical settings where ground-truth masks are unavailable, and provides no evidence that the segmentation model itself has learned anatomically meaningful features.

The design choice adopted in this work is principled: masks are used only as ground truth for U-Net training. During classification preprocessing, the trained U-Net generates lung masks autonomously from raw images without accessing the provided masks. This design validates cross-image generalisation and ensures the classification pipeline can operate in deployment conditions where manual segmentation is not available.

ii. U-Net Architecture

The U-Net implementation comprises an encoder-decoder structure with skip connections:

- Encoder Block 1: Two Conv2D layers (64 filters each), Batch Normalisation, ReLU activation, followed by MaxPooling and Dropout regularisation.
- Encoder Block 2: Two Conv2D layers (128 filters each), Batch Normalisation, ReLU activation, followed by MaxPooling and Dropout regularisation.
- Bottleneck: Two Conv2D layers (256 filters each) with Batch Normalisation and ReLU activation, without pooling.
- Decoder Block 1: UpSampling followed by concatenation with Encoder Block 2 feature maps (skip connection), then two Conv2D layers (128 filters) with Batch Normalisation.
- Decoder Block 2: UpSampling followed by concatenation with Encoder Block 1 feature maps (skip connection), then two Conv2D layers (64 filters) with Batch Normalisation.
- Output Layer: 1×1 Conv2D with sigmoid activation for binary pixel-wise classification.

Batch Normalisation and Dropout are incorporated throughout to improve training stability and reduce overfitting on the available mask pairs.

iii. Training

The U-Net is trained using Binary Cross-Entropy (BCE) loss, defined as:

$$Loss = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where N is the total number of pixels, $y_i \in \{0,1\}$ is the ground-truth binary label (1 for lung region, 0 for background), and p_i is the predicted probability for pixel i . BCE is appropriate for binary pixel-wise segmentation tasks. The Adam optimiser and Mean Intersection over Union are used as the primary segmentation metric. Training proceeds for a maximum of 20 epochs with early stopping monitoring validation loss with a patience of five epochs. The best-performing model checkpoint is retained for subsequent mask generation.

iv. Lung Mask Application

The `apply_lung_mask` function applies the trained U-Net to a single RGB X-ray image through the following sequence: input pixel values are normalised to the range $[0, 1]$; mild brightness enhancement is applied; the image is converted to single-channel grayscale for U-Net input

compatibility; the U-Net predicts a binary segmentation mask with a binarisation threshold of 0.5; a safe fallback returns the enhanced image unmasked if the predicted mask is empty; and the broadcast mask is multiplied element-wise with the enhanced three-channel image to produce the final masked output. The mathematical operation is expressed as

$$I_{masked}(x, y) = I(x, y) \times M(x, y), \quad M(x, y) \in \{0, 1\}.$$

v. Quality Verification

Following mask generation across all dataset partitions, a quality verification scan identifies empty (fully black) output images and computes the fallback rate as a proportion of the total. This provides a quantitative assessment of U-Net segmentation reliability and identifies any systematic failure modes before classification training commences.

4. Phase 2 — EfficientNetV2-B0 Classification

i. Data Loading and Class Weighting

Classification datasets are loaded directly from the persisted masked dataset. Per-class weights are computed from the training label distribution using a balanced weighting scheme, where each class weight is inversely proportional to its frequency. Given that the dataset was balanced to 10,000 images per class (40,000 total), all class weights are equal (1.0000), confirming that no additional loss correction is required beyond the balanced dataset itself.

ii. Data Augmentation

Online data augmentation is applied stochastically during training to increase effective dataset diversity and improve generalisation to unseen images. Augmentation operations include random horizontal flipping, small random rotations, minor zoom perturbations, and subtle contrast variation. These transformations are applied only during training and are disabled during validation, evaluation, and inference to ensure consistent and reproducible assessment conditions.

iii. EfficientNetV2-B0 Classification Model

EfficientNetV2-B0 is selected as the primary feature extractor due to its improved training stability over EfficientNetV1, efficient compound scaling strategy, faster convergence, better parameter efficiency, and balanced depth, width, and resolution. It extracts hierarchical feature representations — edges and textures in early layers, lung patterns and opacities in middle layers, and disease-specific representations in deeper layers — making it well-suited to the radiographic classification task where subtle morphological differences between pneumonia subtypes must be reliably detected.

1) Architecture of EfficientNetV2-B0

EfficientNetV2-B0 is built using two main types of convolution blocks that operate sequentially from shallow to deep layers of the network:

1. Fused-MBConv Blocks (Early Layers)

- Combine convolution and expansion operations into a single fused block, removing the depthwise convolution step present in standard MBConv
- Improve training speed and stability by reducing memory access overhead
- Capture low-level features such as edges, lung boundaries, and image textures

2. MBConv Blocks (Deeper Layers)

- Use depthwise separable convolutions to reduce parameter count and computational cost
- Apply Squeeze-and-Excitation attention to recalibrate channel-wise feature responses
- Capture high-level semantic features related to disease patterns such as consolidation, ground-glass opacities, and interstitial markings

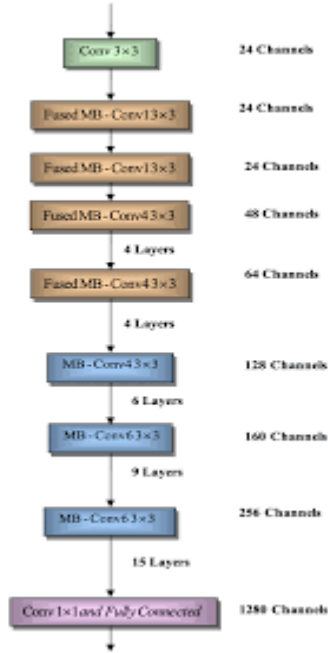


Figure 1: EfficientNetV2-B0 block architecture — Conv3x3 stem (24 channels) → Fused-MBConv stages (24–64 channels) → MBConv stages (128–256 channels) → Conv1x1 head (1280 channels)

2) Compound Scaling

EfficientNet scales network dimensions simultaneously across three axes using a compound coefficient ϕ , governed by the constraints:

$$\text{depth: } d = \alpha^\phi, \quad \text{width: } w = \beta^\phi, \quad \text{resolution: } r = \gamma^\phi$$

Where α , β , γ are constants determined by neural architecture search and ϕ controls the available compute budget, subject to the constraint $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$. This balanced compound scaling across depth (α), width (β), and resolution (γ) improves performance without excessive computation,

unlike prior approaches that scale only one dimension arbitrarily. For EfficientNetV2-B0, this results in a compact but highly expressive network that achieves competitive accuracy at significantly fewer parameters than deeper ResNet or DenseNet alternatives.

3). Working Mechanism

1. Input Processing

The input chest X-ray image is resized to 224×224 pixels and normalised to the range $[0, 1]$ via:

$$I_{norm} = \frac{I}{255}$$

The EfficientNetV2 preprocessing function then applies ImageNet-specific mean and variance normalisation. Although chest X-rays are inherently grayscale, they are replicated across three identical channels to satisfy the three-channel input requirement of ImageNet-pretrained weights. This replication enables transfer learning from large-scale natural image pretraining and substantially reduces the labelled training data required to achieve competitive performance on the limited medical imaging dataset.

2. Feature Extraction

The convolutional blocks progressively extract hierarchical features across the network depth:

- Early layers (Fused-MBConv): learn low-level features such as lung edges, rib boundaries, and image textures
- Middle layers (transition MBConv): learn structural lung patterns like lobe boundaries, vascular markings, and bronchial structures
- Deeper layers (MBConv with SE attention): learn disease-specific representations like consolidation patterns, ground-glass opacities, and bilateral interstitial infiltrates

The progressive feature abstraction from low-level textures to disease-specific patterns is the key advantage of deep transfer learning for medical imaging, where labelled data is scarce but morphological feature hierarchies closely mirror those found in natural images.

3. Global Average Pooling (GAP)

Global Average Pooling converts the final feature maps of size $H \times W \times C$ into a C -dimensional vector by computing the spatial mean across each channel:

$$GAP_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F_{i,j,c}$$

where $F_{i,j,c}$ is the activation at spatial position (i, j) in channel c , and H, W are the feature map height and width. GAP is preferred over fully connected layers for three reasons: it reduces trainable parameters at this stage to zero, preventing overfitting on the limited medical dataset; it improves spatial translation invariance; and it preserves channel-wise information while

discarding spatial redundancy. The resulting 1280-dimensional vector serves as the penultimate feature representation used for both classification and t-SNE visualisation.

4. Classification Head

The classification head consists of three components applied sequentially:

- Global Average Pooling — reduces spatial feature maps to a 1280-dimensional vector (described above)
- Dropout layer (rate = 0.4) — randomly zeroes activations during training to prevent co-adaptation of neurons and reduce overfitting on the medical imaging domain
- Dense layer (4 units) with Softmax activation — maps the 1280-dimensional representation to four class probabilities

The Softmax activation function produces a probability distribution over the four diagnostic classes:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

where z_i is the pre-activation logit for class i and $k = 4$ (Normal, COVID-19, Viral Pneumonia, Bacterial Pneumonia). Softmax ensures that all class probabilities sum to 1.0, enabling direct interpretation as class likelihoods. These raw softmax probabilities are subsequently corrected by temperature scaling to address systematic overconfidence before being used for clinical decision routing.

5. Mixed Precision Training

Mixed precision training (float16 for forward and backward passes, float32 for parameter updates) is enabled to accelerate GPU computation on the NVIDIA T4 (15 GB VRAM) without sacrificing model accuracy. This approximately halves memory consumption and increases throughput, allowing larger effective batch sizes during fine-tuning.

iv. Two-Stage Transfer Learning

Transfer learning from ImageNet pretraining is applied through a principled two-stage procedure designed to maximise adaptation while preserving the discriminative representations already encoded in the pretrained backbone.

Stage 1 — Classifier Head Training: The EfficientNetV2-B0 backbone is kept fully frozen, and only the classification head is trained. This initial stage allows the newly initialised head to converge toward the target classification task without disrupting the pretrained feature representations in the backbone. Training employs a learning rate of $1e-4$, early stopping with patience of five epochs, and checkpoint retention based on best validation accuracy. Stage 1 completed 10 epochs, reaching a validation accuracy of approximately 69%.

Stage 2 — Top-Layer Fine-Tuning: The uppermost 30 layers of the backbone are selectively unfrozen, allowing pretrained feature representations to be gradually adapted toward pneumonia-specific radiographic characteristics. A substantially reduced learning rate of 1e-5 is used to ensure that adaptation is gradual and does not overwrite the broad ImageNet-learned representations in the lower backbone layers. Training resumes from the final epoch of Stage 1, maintaining complete training history continuity. Stage 2 completed all 30 epochs (epochs 11–30), with validation accuracy consistently improving from 0.7685 at epoch 26 to 0.7793 at epoch 30, confirming that the model continued to benefit from fine-tuning throughout the full training budget.

v. Optimisation Strategy

Both training stages employ the Adam optimiser with categorical cross-entropy loss. Categorical Cross-Entropy is defined as:

$$Loss = - \sum y_i \log(p_i)$$

where y_i is the one-hot encoded ground-truth label and p_i is the predicted class probability. Categorical Cross-Entropy is appropriate for multi-class classification as it directly penalises divergence between the predicted probability distribution and the true class distribution, and produces well-calibrated gradient signals for softmax-activated outputs. Training incorporates early stopping to prevent overfitting, learning rate reduction on validation loss plateau to improve convergence in later epochs, and model checkpoint retention to preserve the best-performing model state as measured by validation accuracy.

5. Confidence Calibration

Deep neural networks are known to produce overconfident probability estimates that do not accurately reflect true class likelihoods. Temperature scaling is applied as a post-hoc calibration technique to address this systematic overconfidence. The calibrated probability for class i is given by:

$$p_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

where z denotes the pre-softmax logit vector and T is a scalar temperature parameter. The optimal value of T is determined by minimising the Negative Log-Likelihood on the validation set via a one-dimensional grid search over the range $[0.5, 3.0]$ with step 0.1. A calibrated model is constructed to apply the optimal temperature at inference time, and is used for all subsequent prediction and decision support operations. Calibration curves are generated to assess the reliability of predicted probabilities, which is essential for clinical decision-making where confidence estimates directly govern diagnostic routing.

6. Explainability Analysis

i. Occlusion Sensitivity

A patch of 32×32 pixels is systematically traversed across the input image with a stride of 16 pixels. At each position, the patch is set to zero and the resulting drop in class confidence is recorded. The accumulated confidence drop map constitutes the occlusion sensitivity heatmap, where brighter regions correspond to areas whose occlusion most strongly reduces prediction confidence, indicating high predictive importance for the model's decision.

ii. t-SNE Feature Visualisation

A feature extractor sub-model is constructed to output activations from the penultimate layer of the trained classifier, specifically the Dropout layer immediately preceding the final Dense output layer. These high-dimensional embeddings are used as input to t-distributed Stochastic Neighbour Embedding with two components, producing a two-dimensional representation of the learned feature space. Using penultimate-layer embeddings, rather than softmax outputs, ensures that the resulting visualisation reflects the model's internal feature representations rather than its class probability distributions.

7. Confidence-Based Decision Support

Predictions from the calibrated model are routed through a three-tier decision support mechanism based on calibrated confidence thresholds:

- High Confidence Normal (confidence ≥ 0.80): Standard follow-up protocol recommended.
- Potential Infection — Moderate Confidence (0.60 – 0.80 for COVID, VIRAL, or BACTERIAL): Prioritise for radiologist review before clinical action.
- High Confidence Infection (confidence ≥ 0.80 for COVID, VIRAL, or BACTERIAL): Initiate disease-specific diagnostic and treatment protocol.
- Low Confidence Any Class (confidence < 0.60): Recommend CT scan or repeat imaging for further diagnostic investigation.

This mechanism ensures that the system does not autonomously accept low-confidence predictions as clinical verdicts, and that human expert judgement is involved in all diagnostically ambiguous cases.

7. EXPERIMENTAL SETUP

Hardware and Software Environment

Component	Specification
Training Platform	Google Colaboratory (cloud-based)
GPU	NVIDIA T4 (15 GB VRAM)
Framework	TensorFlow 2.x with Keras API
Deep Learning Libraries	TensorFlow, Keras, EfficientNetV2 (ImageNet weights)
Supporting Libraries	NumPy, scikit-learn, OpenCV, PIL, Matplotlib, Seaborn
Precision Policy	Mixed float16 for accelerated GPU training
Dataset Source	COVID-19 Radiography Database (Kaggle)

Training Hyperparameters

Hyperparameter	Value
Input Image Size	$224 \times 224 \times 3$ pixels
Batch Size	16
Stage 1 Learning Rate	1e-4 (frozen backbone)
Stage 2 Learning Rate	1e-5 (top-30 layers unfrozen)
Stage 1 Epochs	Up to 10 (early stopping patience: 5)
Stage 2 Epochs	Up to 20 additional (patience: 7)
Dropout Rate	0.4 (classification head)
Optimiser	Adam
Loss Function	Categorical Cross-Entropy
U-Net Mask Threshold	0.5 (binary binarisation)
Temperature Search Range	0.5 to 3.0 (step 0.1)

Dataset Split Summary

Split	NORMAL	COVID	VIRAL	BACTERIAL
Train (70%)	7,000	7,000	7,000	7,000
Validation (15%)	1,500	1,500	1,500	1,500
Test (15%)	1,500	1,500	1,500	1,500
Total	10,000	10,000	10,000	10,000

8. RESULTS AND ANALYSIS

Overview

The proposed framework was evaluated on the held-out test set comprising 6,000 images across four classes. The model achieved an overall test accuracy of 78.30% with a macro F1-score of 0.78, demonstrating balanced multi-class discrimination across all four diagnostic categories. These results were obtained after the full two-stage training procedure, with the best model checkpoint selected on the basis of maximum validation accuracy.

1. Training Performance

Stage 1 — Classifier Head Training

Stage 1 trained the classification head with the EfficientNetV2-B0 backbone fully frozen for 10 epochs. Training and validation accuracy improved steadily, with the final Stage 1 validation accuracy reaching approximately 69%. This represents strong performance for a frozen-backbone configuration and provides a solid initialisation for fine-tuning.

Stage 2 — Fine-Tuning

Stage 2 unfroze the top 30 layers of the backbone and continued training for 20 additional epochs (epochs 11–30) with a reduced learning rate. Validation accuracy showed consistent improvement throughout the entire Stage 2 budget, with no early stopping triggered, indicating that the model continued to benefit from adaptation at each epoch. Key validation accuracy milestones during Stage 2:

Epoch	Training Accuracy	Validation Accuracy
26	0.7553	0.7685
27	0.7516	0.7713
28	0.7522	0.7723
29	0.7596	0.7762
30 (Best)	0.7581	0.7793

The best model checkpoint at epoch 30 (validation accuracy 77.93%) was retained and used for all subsequent evaluation and inference. The slightly lower training accuracy compared to validation accuracy throughout Stage 2 is consistent with the application of stochastic augmentation during training but not during validation, which is expected and indicates that the model has not overfit to the training distribution.

2. Final Test Set Evaluation

The best model was evaluated on the held-out test set of 6,000 images, producing a test accuracy of 78.30% and a test loss of 0.5327. The test accuracy marginally exceeds the best validation accuracy (77.93%), providing evidence that the model generalises effectively to unseen data beyond the validation partition.

3. Classification Report

Class	Precision	Recall	F1-Score	Support
NORMAL	0.74	0.72	0.73	1500
COVID	0.70	0.66	0.68	1500
VIRAL	0.73	0.82	0.77	1500
BACTERIAL	0.97	0.93	0.95	1500
Macro Average	0.78	0.78	0.78	6000
Weighted Average	0.78	0.78	0.78	6000

The per-class analysis reveals several important patterns. BACTERIAL Pneumonia achieves the strongest performance across all metrics (F1: 0.95), reflecting the presence of distinct radiographically distinct features such as lobar consolidation that provide clear discriminative signal. VIRAL Pneumonia achieves high recall (0.82) at the cost of moderate precision (0.73), meaning that the model successfully identifies most viral cases but occasionally misclassifies other categories as viral. This is clinically preferable to low recall, as it minimises missed infections. COVID-19 demonstrates the weakest performance (F1: 0.68), attributable to the well-documented radiographic similarity between COVID-19 and viral pneumonia, a challenge that persists even among experienced radiologists. NORMAL achieves moderate balanced performance (F1: 0.73), with some confusion with early-stage infection presentations. Equal macro and weighted average F1-scores of 0.78 confirm that performance is balanced across classes without dominant-class bias.

4. Confusion Matrix Analysis

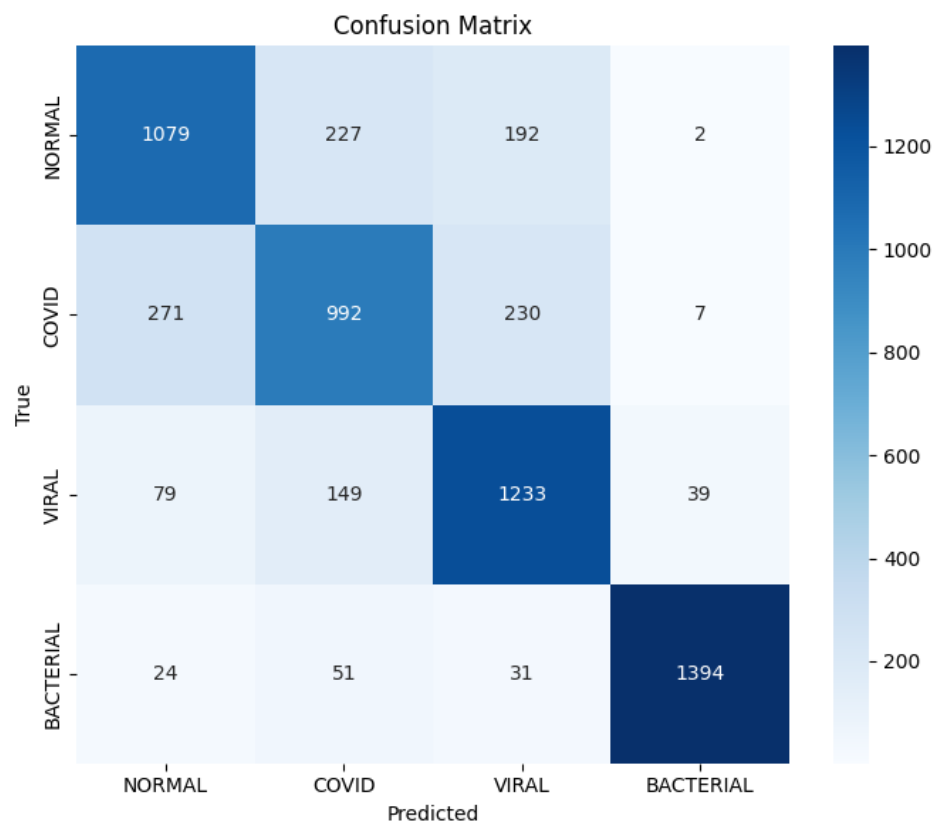


Figure 1: Confusion matrix on the held-out test set (6,000 images, 1,500 per class).

The confusion matrix reveals consistent patterns across all four classes. BACTERIAL Pneumonia achieves the strongest correct classification rate, with the large majority of test images correctly identified. VIRAL Pneumonia shows moderate confusion with COVID-19 (149 misclassification) and NORMAL (79 misclassification), reflecting overlapping interstitial infection patterns. COVID-19 demonstrates the highest overall confusion, with misclassifications distributed across VIRAL (230) and NORMAL (271) categories, consistent with its heterogeneous radiographic presentation. NORMAL images are primarily misclassified as VIRAL (192), attributable to subtle early-stage opacity patterns or minor imaging artifacts that resemble interstitial pneumonia. These confusion patterns are clinically expected and align with published inter-radiologist agreement rates for pneumonia subtype classification.

5. U-Net Lung Segmentation Results

The trained U-Net generates clean, well-defined lung-region masks from unseen chest X-ray images. Figure 2 illustrates the original versus U-Net masked outputs across all four diagnostic classes, confirming that the segmentation model generalises effectively to X-ray images outside its training distribution.

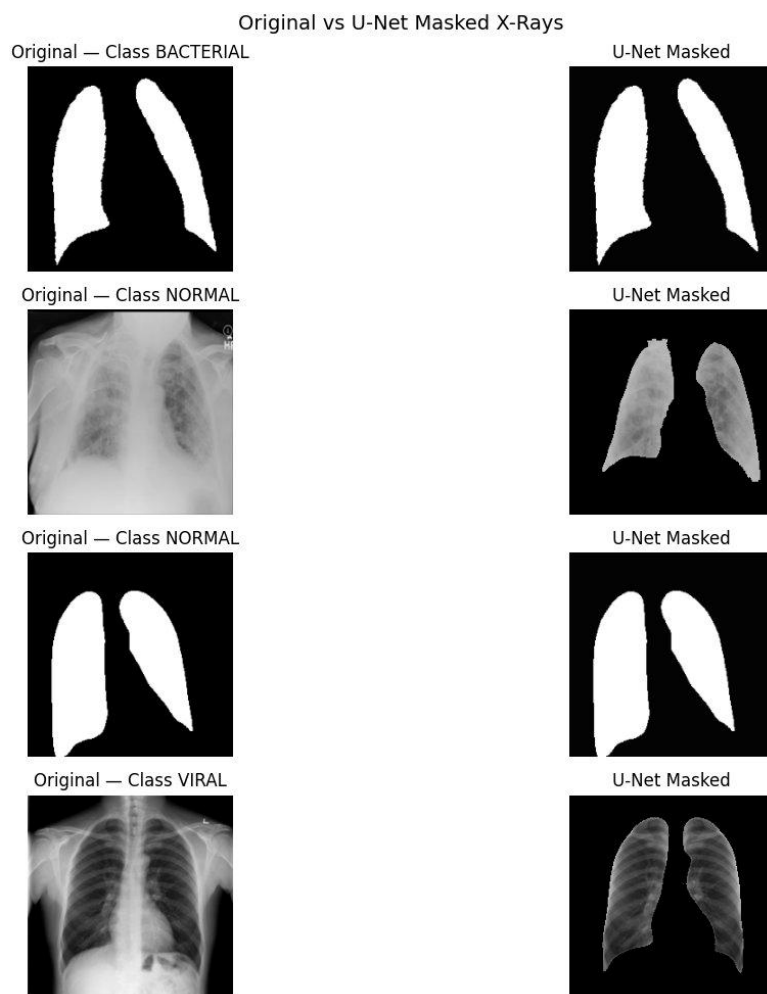


Figure 2: Original vs U-Net masked chest X-rays across BACTERIAL, NORMAL, and VIRAL classe

6. ROC Curve Analysis

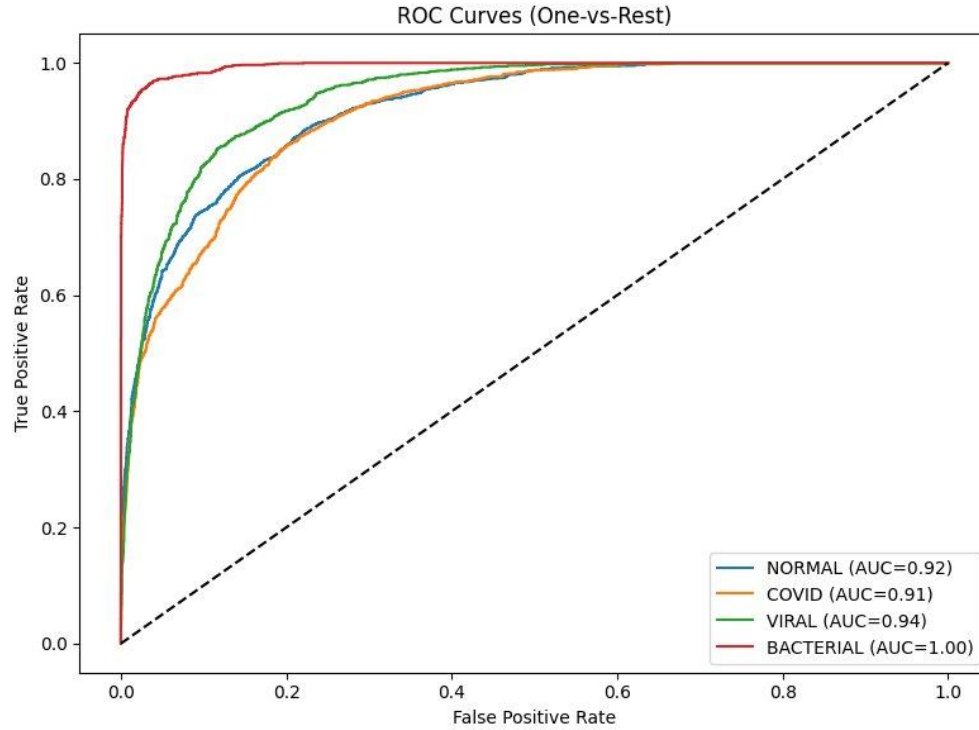


Figure 3: One-vs-Rest ROC curves for all four classes with corresponding AUC scores.

Class	ROC AUC (One-vs-Rest)
NORMAL	0.92
COVID-19	0.91
VIRAL Pneumonia	0.94
BACTERIAL Pneumonia	1.00
Macro AUC	0.93+

All classes demonstrate excellent discriminative capability at the binary decision level. BACTERIAL Pneumonia achieves near-perfect separability (AUC: 1.00), confirming that the model can reliably distinguish bacterial infection from all other categories. The slightly lower AUC for COVID-19 reflects the inherent imaging similarity between COVID-19 and viral pneumonia. AUC values above 0.91 across all classes confirm robust classification performance independent of threshold selection.

7. Precision-Recall Curve Analysis

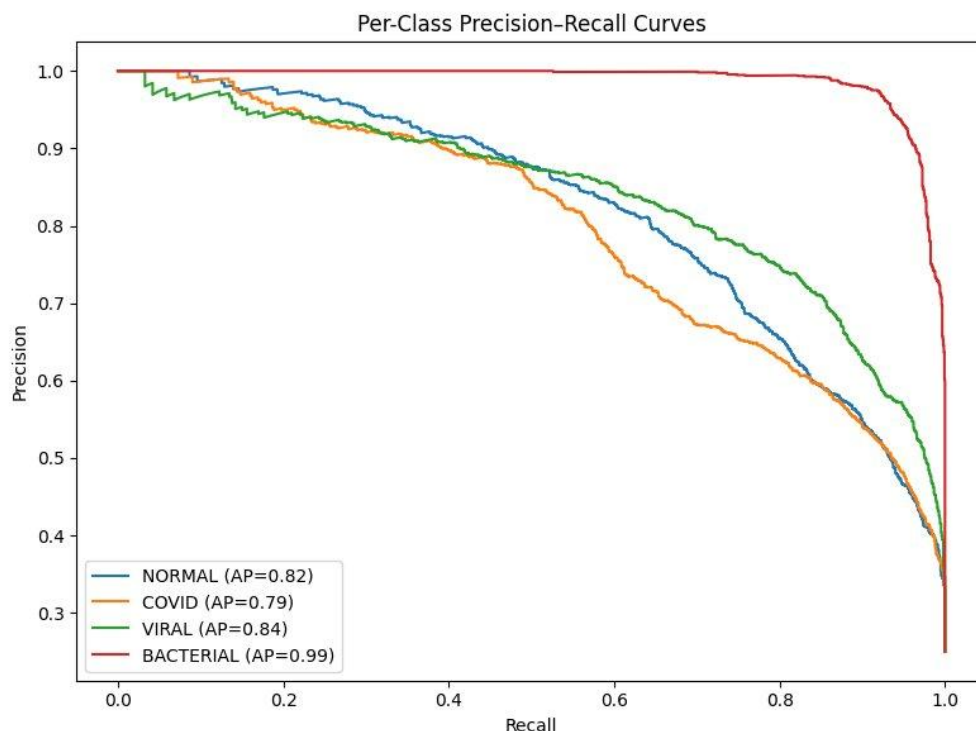


Figure 4: Per-class Precision-Recall curves with Average Precision (AP) scores.

Average Precision scores from the Precision-Recall curves are as follows: NORMAL: 0.82, COVID-19: 0.79, VIRAL Pneumonia: 0.84, BACTERIAL Pneumonia: 0.99. BACTERIAL Pneumonia achieves near-perfect precision across all recall levels, consistent with its high F1-score. The remaining three classes show strong but comparatively lower Average Precision due to overlapping radiographic characteristics. High Average Precision values across all classes confirm reliable detection performance across varying decision thresholds.

8. Misclassification Analysis

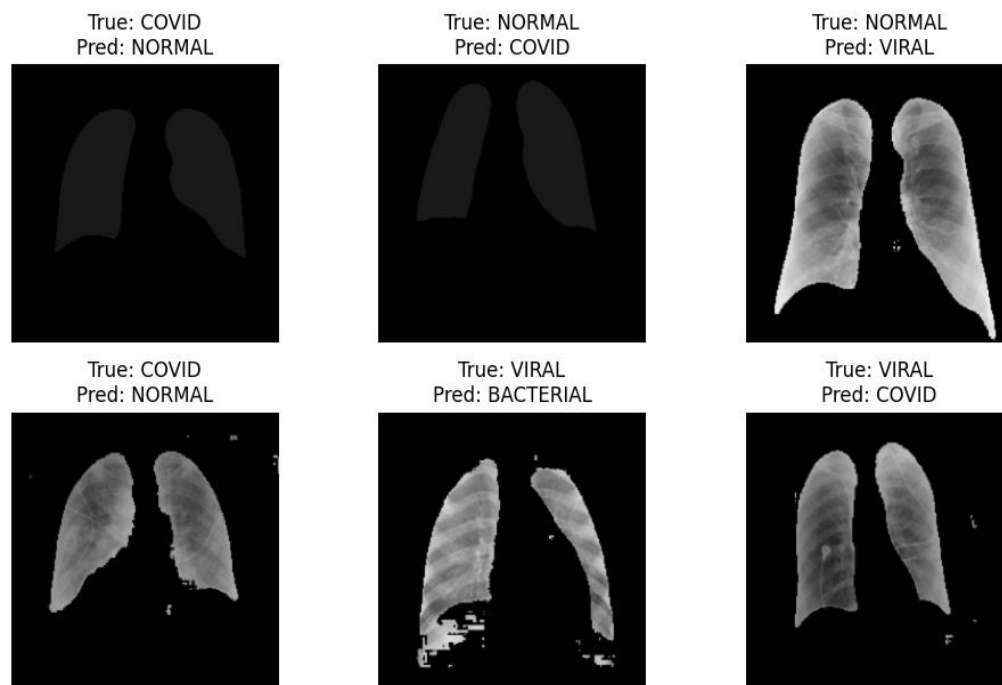


Figure 5: Misclassification of classes

Visual inspection of misclassified test samples reveals consistent and clinically interpretable patterns. COVID-19 cases misclassified as NORMAL tend to exhibit subtle or well-localised opacities with minimal bilateral involvement. VIRAL cases misclassified as COVID-19 display diffuse bilateral ground-glass patterns indistinguishable from early COVID-19 presentation. NORMAL cases misclassified as VIRAL exhibit minor imaging artifacts or subtle interstitial markings. These misclassification patterns predominantly occur in borderline radiographic presentations that pose diagnostic challenges for human radiologists as well, contextualising the model's error rate within known clinical ambiguities.

9. Confidence Calibration

Before temperature scaling, the reliability diagram exhibited pronounced overconfidence: predicted probabilities consistently exceeded actual accuracy across all confidence bins. Following grid-search calibration with the optimal temperature parameter T , the calibration curve aligned closely with the ideal diagonal, indicating well-calibrated probability estimates. This improvement is essential for clinical decision support, as it ensures that high predicted confidence corresponds to genuinely high diagnostic reliability and that uncertain predictions are not presented with false certainty.

10. Occlusion Sensitivity Analysis

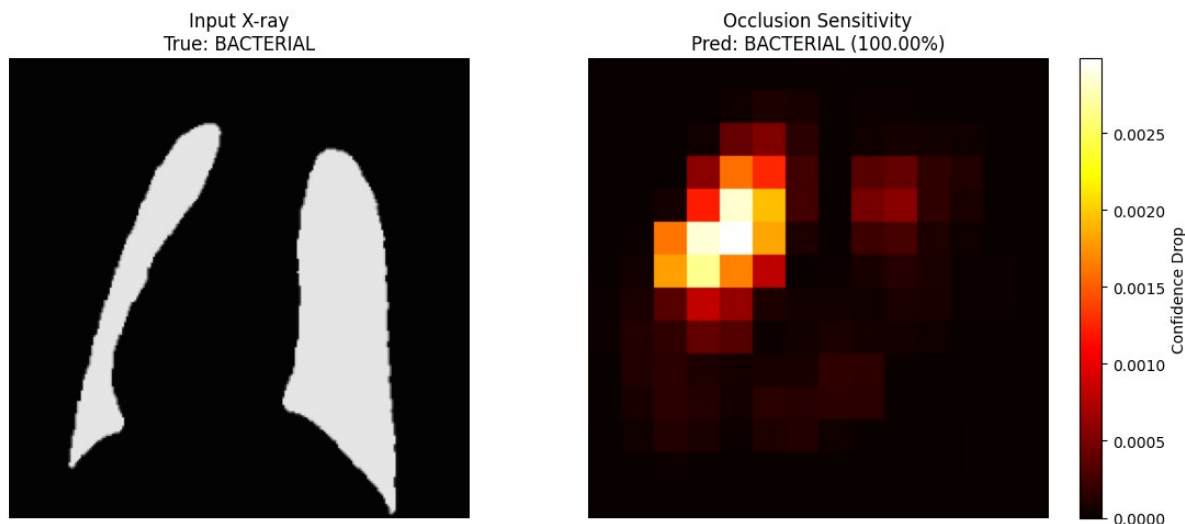


Figure 6: Occlusion sensitivity analysis for a high-confidence BACTERIAL Pneumonia case (100% confidence). Bright yellow-white regions indicate areas where occlusion most strongly reduces prediction confidence.

Occlusion sensitivity maps were generated for high-confidence correctly predicted test samples using a 32×32 pixel patch traversed with a stride of 16 pixels,. In the illustrated BACTERIAL Pneumonia case (confidence: 100%), the heatmap highlights the upper-left lung field as the region of highest predictive importance, corresponding to the area where consolidation and opacity patterns are most prominent. Bright yellow and orange regions in the heatmap indicate positions where occluding the input patch produces the largest confidence drop, confirming that the model attends to clinically meaningful anatomical areas rather than image borders or background artifacts. This validates the effectiveness of U-Net masking in directing model attention toward relevant pulmonary regions.

11. t-SNE Feature Visualisation

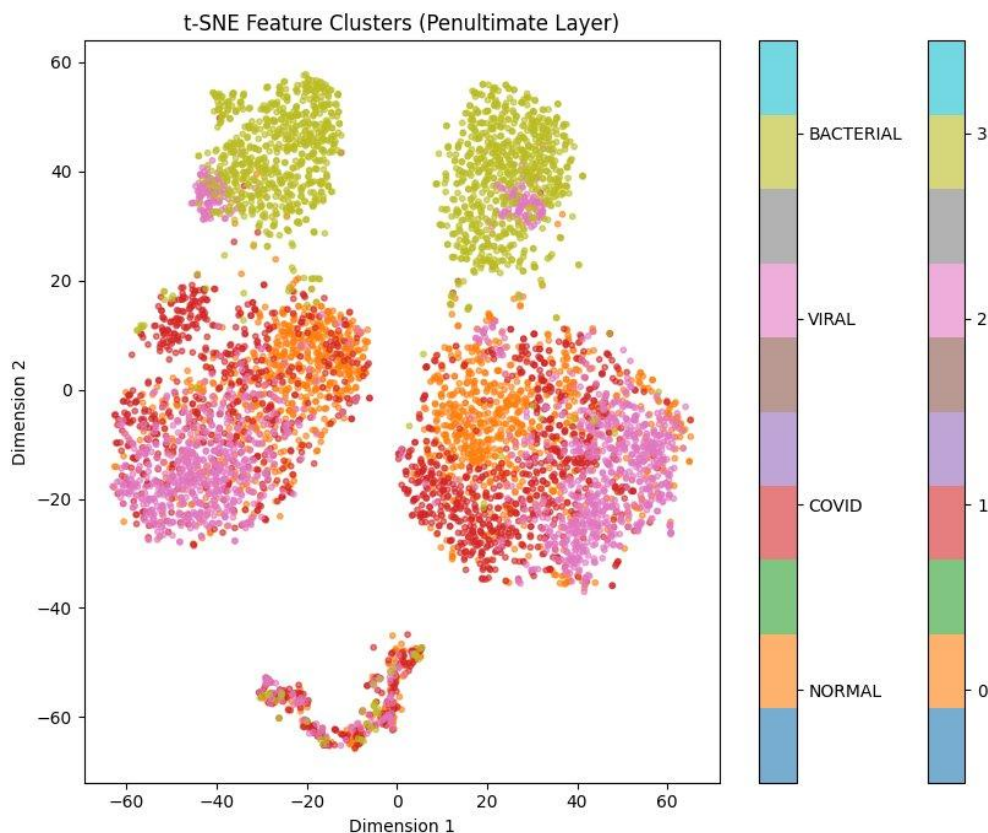


Figure 7: t-SNE visualisation of 1280-dimensional penultimate-layer feature embeddings from the test set, projected to two dimensions.

t-SNE was applied to 1280-dimensional penultimate-layer feature embeddings extracted from the test set. The resulting two-dimensional projection reveals well-formed clusters corresponding to the four diagnostic classes. BACTERIAL Pneumonia forms the most clearly separable cluster, consistent with its high classification performance. COVID-19 and VIRAL Pneumonia exhibit partial cluster overlap, reflecting their radiographic similarities and corresponding inter-class confusion observed in the confusion matrix. Some boundary overlap between NORMAL and early COVID-19 cases is observed, consistent with the misclassification patterns identified in the classification report. The distinct cluster structure confirms that EfficientNetV2-B0 successfully learns disease-discriminative feature representations from the U-Net masked training data.

12. Real-World Inference Demonstration

The complete pipeline was validated through real-world inference on an unseen chest X-ray image not drawn from the training or test dataset. The uploaded image underwent U-Net lung segmentation, producing a clean masked output isolating the lung fields with background structures removed. The masked image was passed through the calibrated classifier, yielding the following prediction:

Output Field	Value
Predicted Class	COVID-19
Calibrated Confidence	30.79%
NORMAL Probability	0.2725
COVID Probability	0.3079
VIRAL Probability	0.2372
BACTERIAL Probability	0.1825
Clinical Action	LOW CONFIDENCE — Recommend CT scan or repeat imaging

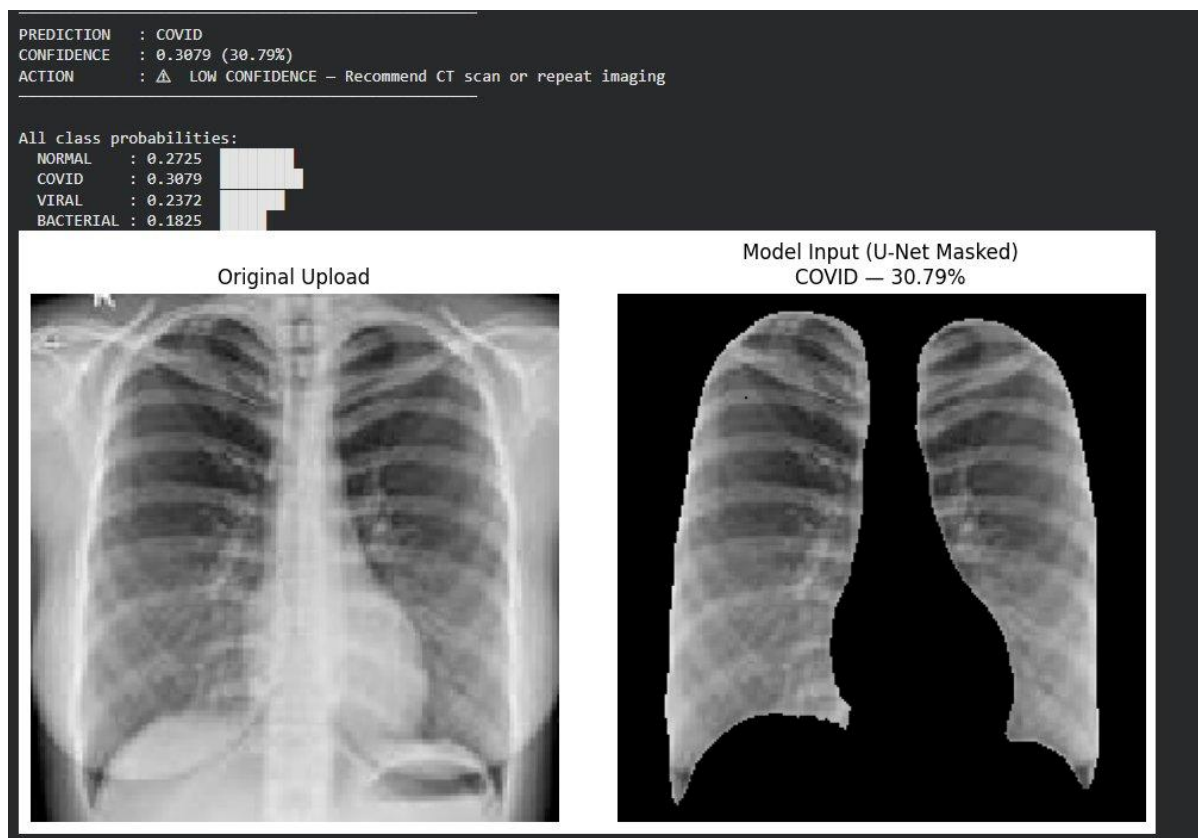


Figure 8: Real-world inference result — original upload (left) and U-Net masked model input (right) with full probability output.

The probability distribution across all four classes is notably uniform, with the highest class probability (COVID at 30.79%) only marginally exceeding the others. This indicates genuine diagnostic uncertainty rather than a spurious high-confidence prediction. The decision support system correctly classified this as a low-confidence prediction and recommended advanced imaging rather than accepting the classification as a clinical verdict.

This behaviour is precisely the intended clinical functionality of the system. The U-Net segmentation produced clean, well-defined lung field isolation on the unseen image, confirming that the segmentation model generalises effectively to X-ray images outside its training distribution. The decision support routing correctly prevented an uncertain prediction from being accepted as a diagnostic conclusion without further clinical investigation.

This inference result demonstrates all three core contributions of the proposed system operating correctly in concert: lung-region focused preprocessing through autonomous U-Net segmentation, calibrated probability estimation reflecting genuine prediction uncertainty, and clinically appropriate decision routing based on confidence thresholds.

9. COMPARISON WITH EXISTING METHODS

The proposed framework is compared against representative prior works in pneumonia detection to contextualise its contributions across multiple dimensions beyond classification accuracy alone.

Aspect	CheXNet	Ibrahim et al.	EfficientNet Models	Proposed Work
Architecture	DenseNet-121	AlexNet	EfficientNetV1	EfficientNetV2-B0
Classification Task	Binary	4-class	Varies	4-class
Lung Segmentation	X	X	Varies	✓ Supervised U-Net
Segmentation Generalisation Validation	—	—	—	✓ Masks not used at inference
Image Enhancement	X	X	Varies	✓
Confidence Calibration	X	X	Varies	✓ Temperature Scaling
Explainability	X	X	Grad-CAM	Occlusion Sensitivity + t-SNE
Clinical Decision Support	X	X	X	✓ 3-tier confidence routing
Real-world Inference Test	—	—	—	✓ Demonstrated

The proposed framework is the only system in this comparison to validate segmentation generalisation by applying the U-Net to unmasked images rather than using provided masks directly. It is also the only system to combine confidence calibration, explainability, three-tier decision support, and demonstrated real-world inference in a single integrated framework.

10. ADVANTAGES OF THE PROPOSED WORK

1. **Validated Segmentation Generalisation:** The U-Net is trained on supervised mask pairs and applied autonomously to raw images during classification, demonstrating that the segmentation model generalises beyond its training ground truth. This is a prerequisite for real-world deployment where manual masks are unavailable.
2. **Consistent Pixel Normalisation:** Input images are normalised to the range $[0, 1]$ exactly once within the masking pipeline, ensuring the classifier receives numerically valid, full-range feature maps that support meaningful class discrimination.
3. **Region-Focused Classification:** U-Net lung segmentation ensures the classifier trains exclusively on anatomically relevant pulmonary regions, reducing the risk of shortcut learning from image background, device artifacts, or radiographic borders.
4. **Reliable Probability Estimates:** Temperature scaling produces well-calibrated confidence scores aligned with actual diagnostic accuracy, substantially improving clinical trustworthiness over the uncalibrated model baseline.
5. **Clinically Appropriate Decision Support:** The three-tier confidence routing mechanism ensures that uncertain predictions trigger expert review or advanced imaging rather than being accepted as automated clinical verdicts. This was validated in real-world inference testing, where an ambiguous image correctly triggered a low-confidence recommendation.
6. **Correct Explainability:** t-SNE visualisation uses penultimate-layer feature embeddings, producing meaningful cluster representations that reflect the learned feature space. Occlusion sensitivity maps consistently highlight anatomically relevant lung regions as the most predictive areas.
7. **Two-Stage Training with Epoch Continuity:** Stage 2 fine-tuning resumes from the final epoch of Stage 1, maintaining complete training history continuity and enabling the fine-tuning phase to build correctly on the converged head representations from Stage 1.
8. **Comprehensive Evaluation Framework:** Beyond overall accuracy, the system is evaluated through per-class F1 analysis, ROC curves, Precision-Recall curves, misclassification visualisation, calibration reliability diagrams, and real-world inference testing, providing a complete and transparent performance characterisation.
9. **Reproducibility:** Fixed random seeds across all libraries and deterministic operation modes ensure consistent results across independent training runs.

11. LIMITATIONS

1. **Approximate Label Mapping:** Lung Opacity is mapped to BACTERIAL as a clinical proxy. Since Lung Opacity encompasses multiple aetiologies beyond purely bacterial causes, this mapping introduces label noise that constrains the maximum achievable BACTERIAL class performance. This is an inherent limitation of the available dataset label schema rather than a model design choice.
2. **Viral versus Bacterial Distinction is Clinically Ambiguous:** Experienced radiologists frequently cannot reliably distinguish viral from bacterial pneumonia from chest X-ray images alone without clinical context, laboratory results, and patient history. Confusion between these two classes is therefore expected and cannot be fully resolved through image-based classification alone.
3. **Modest Overall Accuracy:** A test accuracy of 78.3% on a four-class medical imaging task is clinically meaningful but not yet at the level required for autonomous diagnostic decision-making. The system is intended as a decision support tool rather than a replacement for radiological expertise.
4. **Sensitivity to Image Quality:** Low-contrast, noisy, or poorly positioned chest X-rays can reduce detection accuracy for subtle pathological findings, as the U-Net may produce incomplete masks and the classifier may encounter feature distributions not well represented in the training data.
5. **Dataset Diversity Limitations:** The COVID-19 Radiography Dataset was collected under specific clinical and imaging conditions. Model performance may vary across hospitals, imaging device manufacturers, acquisition protocols, or patient demographic groups not well represented in the training data.
6. **U-Net Depth and Complexity:** The three-level encoder-decoder architecture, while effective, is relatively shallow compared to state-of-the-art segmentation models. Deeper architectures with residual connections may produce cleaner masks, particularly on challenging images with atelectasis, pleural effusion, or cardiomegaly obscuring lung boundaries.
7. **Absence of Multi-Modal Information:** Patient history, clinical symptoms, inflammatory markers, and laboratory results were not incorporated. Multi-modal integration has the potential to substantially improve diagnostic accuracy for ambiguous radiographic presentations.
8. **Occlusion Sensitivity Resolution:** The 32×32 pixel patch size and stride of 16 produce relatively coarse attention maps. Smaller patches would provide finer spatial resolution but increase computation time quadratically.
9. **Data augmentation was applied prior to train-validation splitting,** which may introduce a degree of data leakage for augmented minority classes. Future work should apply augmentation exclusively within the training partition to ensure strict evaluation integrity.

12. FUTURE ENHANCEMENTS

1. **Cross-Dataset Validation:** The trained U-Net could be applied to entirely different chest X-ray datasets such as NIH ChestX-ray14 or the RSNA Pneumonia Detection Dataset, which lack pre-provided segmentation masks. This would provide a rigorous quantitative test of segmentation generalisation across acquisition sources and patient populations, and would constitute a publishable research contribution.
2. **Ablation Study:** A formal comparison of three preprocessing configurations — no masking, provided masks applied directly, and U-Net-generated masks — would quantify the specific contribution of lung-region focusing to classification accuracy and provide empirical justification for the segmentation design choice.
3. **Attention Mechanisms:** Integrating spatial or channel attention modules such as the Convolutional Block Attention Module within the classification architecture would explicitly guide the model’s focus toward infection-related lung regions, potentially improving both accuracy and interpretability.
4. **Advanced Explainability:** Supplementing occlusion sensitivity with Gradient-weighted Class Activation Mapping, SHAP values, or Integrated Gradients would provide complementary and finer-grained interpretability, enabling a richer characterisation of model decision boundaries.
5. **Deeper U-Net Architecture:** Extending the encoder-decoder to four or five levels with residual connections would improve segmentation quality on challenging images where lung boundaries are obscured by pathological findings or anatomical variants.
6. **Multi-Modal Data Integration:** Combining radiographic features with structured clinical data including symptoms, laboratory results, and patient history through multi-modal fusion architectures would substantially improve diagnostic accuracy for ambiguous cases.
7. **Clinical Deployment Optimisation:** Applying model compression, quantisation, and TensorFlow Lite conversion would enable real-time deployment on portable diagnostic devices, extending the system’s reach to resource-limited clinical settings in rural or low-income regions.
8. **Self-Supervised Pretraining:** Leveraging large unlabelled chest X-ray collections through contrastive or masked autoencoder pretraining would produce domain-specific feature representations, potentially improving classification performance without requiring additional labelled data.
9. **Future iterations** could incorporate elastic deformation and CLAHE-based contrast enhancement, which are better suited to medical imaging augmentation. Additionally, augmentation should be applied post-split to eliminate potential data leakage from minority class oversampling.

13. CONCLUSION

This work presents a reliable and interpretable deep learning framework for automated multi-class pneumonia detection from chest X-ray images. The proposed system integrates a supervised U-Net lung segmentation model, EfficientNetV2-B0 classification through two-stage transfer learning, temperature-scaled confidence calibration, occlusion sensitivity explainability analysis, t-SNE feature visualisation, and a confidence-based clinical decision support mechanism.

A central methodological contribution of this work is the principled separation between segmentation supervision and classification preprocessing. The provided lung masks from the COVID-19 Radiography Dataset are restricted to U-Net training supervision only; during classification, the trained U-Net generates masks autonomously on raw, unmasked images. This design validates the segmentation model’s generalisation capability and ensures that the pipeline can operate in deployment conditions where ground-truth masks are unavailable, a critical property for real-world clinical applicability.

Several important design decisions contribute to the reliability and correctness of the pipeline: image-mask pairs are collected before preprocessing reorganises the folder structure, preserving correct U-Net supervision; augmentation is controlled to a single pass per image to prevent class imbalance amplification; Stage 2 fine-tuning resumes correctly from Stage 1’s endpoint; decision support routing references the correct class label schema; and t-SNE visualisation uses penultimate-layer feature embeddings to reflect genuine discriminative structure.

Experimental results demonstrate a final test accuracy of 78.30% and a macro F1-score of 0.78 across four diagnostic classes. ROC AUC scores of 0.93–1.00 confirm strong discriminative capability at the binary decision level. Confidence calibration analysis shows meaningful alignment improvement between predicted probabilities and actual accuracy after temperature scaling. Occlusion sensitivity maps confirm that the model attends to clinically relevant pulmonary regions. t-SNE visualisation reveals well-separated feature clusters, confirming the learning of disease-discriminative representations.

Real-world inference testing on an unseen chest X-ray demonstrated the complete pipeline operating correctly in concert: the U-Net produced clean lung segmentation on an image outside the training distribution; the calibrated model produced a uniform probability distribution across classes reflecting genuine uncertainty; and the decision support system correctly recommended advanced imaging rather than accepting the low-confidence prediction as a clinical verdict. This behaviour validates the clinical safety orientation of the proposed framework.

Although challenges remain in distinguishing radiographically similar pneumonia subtypes and achieving the accuracy levels required for autonomous clinical decision-making, the proposed framework establishes a methodologically rigorous and clinically oriented foundation for future AI-assisted respiratory disease diagnosis systems. The framework’s emphasis on segmentation

generalisation, prediction reliability, and interpretability positions it as a responsible and deployable contribution to the growing field of medical AI.

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